Pain Medicine, 22(7), 2021, 1485–1495
doi: 10.1093/pm/pnab098
Advance Access Publication Date: 13 March 2021
Original Research Article

OXFORD

INTERVENTIONAL PAIN & SPINE MEDICINE SECTION

Simplified Universal Grading of Lumbar Spine MRI Degenerative Findings: Inter-Reader Agreement of Non-Radiologist Spine Experts

Nityanand Miskin, MD,* Zacharia Isaac, MD,† Yi Lu, MD, PhD,‡ Melvin C. Makhni, MD,§ Danielle L. Sarno, MD,†,‡ Timothy R. Smith, MD, PhD, MPH,‡ Jay M. Zampini, MD,§ and Jacob C. Mandell, MD*

*Department of Radiology; †Department of Physical Medicine and Rehabilitation; ‡Department of Neurosurgery; and §Department of Orthopaedic Surgery, Brigham and Women's Hospital, Harvard Medical School, Boston, Massachusetts, USA

Correspondence to: Nityanand Miskin, MD, 75 Francis Street, Department of Radiology, Boston, MA 02115, USA. Tel: 732-447-6864; Fax: (617) 264-5250; E-mail: nmiskin@bwh.harvard.edu.

Disclosure and conflicts of interest: The authors declare no conflict of interest or sources of funding.

Received on 15 January 2021; revised on 23 February 2021; Accepted on 9 March 2021

Abstract

Objective. 1) To describe a simplified multidisciplinary grading system for the most clinically relevant lumbar spine degenerative changes. 2) To measure the inter-reader variability among non-radiologist spine experts in their use of the classification system for interpretation of a consecutive series of lumbar spine magnetic resonance imaging (MRI) examinations. **Methods**. ATS multidisciplinary and collaborative standardized grading of spinal stenosis, foraminal stenosis, lateral recess stenosis, and facet arthropathy was developed. Our institution's picture archiving and communication system was searched for 50 consecutive patients who underwent non-contrast MRI of the lumbar spine for chronic back pain, radiculopathy, or symptoms of spinal stenosis. Three fellowship-trained spine subspecialists from neurosurgery, orthopedic surgery, and physiatry interpreted the 50 exams using the classification at the L4–L5 and L5–S1 levels. Inter-reader agreement was assessed with Cohen's kappa coefficient. **Results**. For spinal stenosis, the readers demonstrated substantial agreement ($\kappa = 0.702$). For foraminal stenosis and facet arthropathy, the three readers demonstrated moderate agreement ($\kappa = 0.544$, and 0.557, respectively). For lateral recess stenosis, there was fair agreement ($\kappa = 0.323$). **Conclusions**. A simplified universal grading system of lumbar spine MRI degenerative findings is newly described. Use of this multidisciplinary grading system in the assessment of clinically relevant degenerative changes revealed moderate to substantial agreement among non-radiologist spine physicians. This standardized grading system could serve as a foundation for interdisciplinary communication.

Key Words: Spine; Low Back Pain; Lumbar; Degenerative Disc Disease; Magnetic Resonance Imaging; Spinal Stenosis; Facet Joint

Introduction

Low back pain, radiculopathy, and neurogenic claudication together comprise one of the most common reasons for seeking clinical care [1–3] and have a tremendous societal burden [4, 5]. Imaging patients with low back pain continues to be a challenging issue. Prior work has shown that imaging often identifies abnormalities in asymptomatic patients [6–10], and imaging findings may not predict the future development of back pain [11]. Despite these issues, as recommended by the guidelines of the American College of Physicians and the American Pain Society, magnetic resonance imaging (MRI) has a key role for patients with persistent low back and leg pain who are potential candidates for treatment procedures, including injections and surgery [12]. The fundamental principal of interventional treatment of lumbar spine disorders is to target the specific site(s) of pathology that correlate with clinical symptoms, and MRI is the primary

© The Author(s) 2021. Published by Oxford University Press on behalf of the International Life Sciences Institute.
All rights reserved. For permissions, please e-mail: journals.permissions@oup.com.

Downloaded from https://academic.oup.com/painmedicine/article/22/7/1485/6169540 by Harvard Law School Library user on 11 November 2021

modality for evaluation. Symptoms may be due to nerve impingement from spinal stenosis, foraminal stenosis, or lateral recess stenosis, and facet arthropathy is an important contributor to both axial low back pain and mechanical neural impingement from the mass effect of osteophytes or ligamentum flavum hypertrophy. Despite the importance of MRI in guiding clinical management, multiple studies have demonstrated substantial inter-reader variability in interpretation between radiologists [13–17], between radiology subspecialties [18], and between treating specialties, including orthopedics, neurosurgery, and pain management [15, 16, 19]. This variability may introduce ambiguity into the clinical decision-making process, lead to discrepancies between imaging and clinical symptoms, and impact insurance reimbursement. Therefore, efforts to improve inter-reader agreement may be helpful for clinical care.

Multiple parameters of lumbar spine degeneration have been described, including the degree of spinal and foraminal stenoses [20–24], Modic endplate changes [25, 26], facet arthropathy [27, 28], lateral recess stenosis [20, 29], and grading of disc degeneration [30–32]. However, some of these classifications can be cumbersome to utilize, and the majority of these efforts have focused on a single parameter, such as spinal stenosis or foraminal stenosis. To our knowledge, a universal classification system has not yet been described. The need for a universal grading system for degenerative spine MRI interpretation became evident from our institution's multidisciplinary spine conference, with regular and valuable discussion between expert spine physicians from radiology, orthopedic surgery, neurosurgery, and physiatry. We found that certain clinically important parameters were described with variable frequency and reliability. Therefore, the purpose of the present study is to describe a simplified classification for the most clinically relevant lumbar spine degenerative changes. The secondary purpose of this study is to investigate the inter-reader variability among non-radiologist spine experts when they use the classification to interpret lumbar spine MRIs.

Methods

Simplified Universal Classification Framework

The grading system was developed as a multidisciplinary collaboration with experienced, fellowship-trained spine physicians from orthopedic surgery, neurosurgery, physiatry, and radiology. The collaborative group included two spine neurosurgeons with 5 and 7 years of experience, two spine orthopedic surgeons with 2 and 10 years of experience, two physiatrists with 4 and 18 years of experience, and one musculoskeletal radiologist with 6 years of experience (the senior author of this study).

The findings determined to be most clinically relevant for interventional management included spinal stenosis, foraminal stenosis, lateral recess stenosis, and facet arthropathy. Bullet-point descriptions and line-art schematics illustrating each degree of severity were created by the senior author, with iterative feedback from the multidisciplinary group. After the group agreed upon the classification, representative MRIs were selected (separate from the cases used to determine inter-reader agreement) by the senior and first authors in consensus for each grade of each classification. The schematic diagrams and MRIs were combined into a Microsoft PowerPoint file (Microsoft Corp., Redmond, WA, USA), and the schematic illustrations and bullet-point text were condensed into a single-page summary PDF (Adobe, Mountainview, CA, USA). Additionally, the senior author recorded a 19-minute video presenting the classification.

The final classification is as follows:

Spinal Stenosis

Spinal stenosis was assessed with a modified grading system from Schizas et al. [33]. Schizas described seven gradations of spinal stenosis (eight total including normal), where grades A1 through A4 represent progressive degrees of mild spinal stenosis with cerebrospinal fluid (CSF) remaining clearly visible within the dural sac but with inhomogeneous distribution. Grade B in the Shizas classification represents moderate spinal stenosis with persistently visible CSF, grade C represents severe spinal stenosis with effacement of CSF, and grade D represents extreme stenosis with effacement of epidural fat.

Our spinal stenosis classification (Figure 1) simplifies Schizas' classification into normal, mild, moderate, and severe, with intermediate grades of mild–moderate and moderate–severe allowed but not explicitly described. Assessment was performed on axial T2-weighted images, as follows:

Normal: Nerve roots are freely distributed, without crowding. The anterior margin of the thecal sac is flat or convex.

Mild spinal stenosis: There is slight crowding of the nerve roots. The anterior margin of the thecal sac is flat or slightly concave. The nerve roots remain distinguishable from the CSF.

Moderate spinal stenosis: There is crowding of the nerve roots, resulting in a homogeneously "speckled" appearance of CSF interspersed with nerve roots. The anterior margin of the thecal sac is concave.

Severe spinal stenosis: There is complete effacement of CSF, resulting in nerve roots' being not individually distinguishable. The anterior margin of the thecal sac is concave or not discernable.

Foraminal Stenosis

Foraminal stenosis was assessed with a modified grading system developed by Lee et al. [24]. Lee described a four-point scale of severity, with grade 0 being normal, grade 1 representing mild foraminal stenosis with perineural fat effacement in two directions, grade 2 representing moderate

Downloaded from https://academic.oup.com/painmedicine/article/22/7/1485/6169540 by Harvard Law School Library user on 11 November 2021

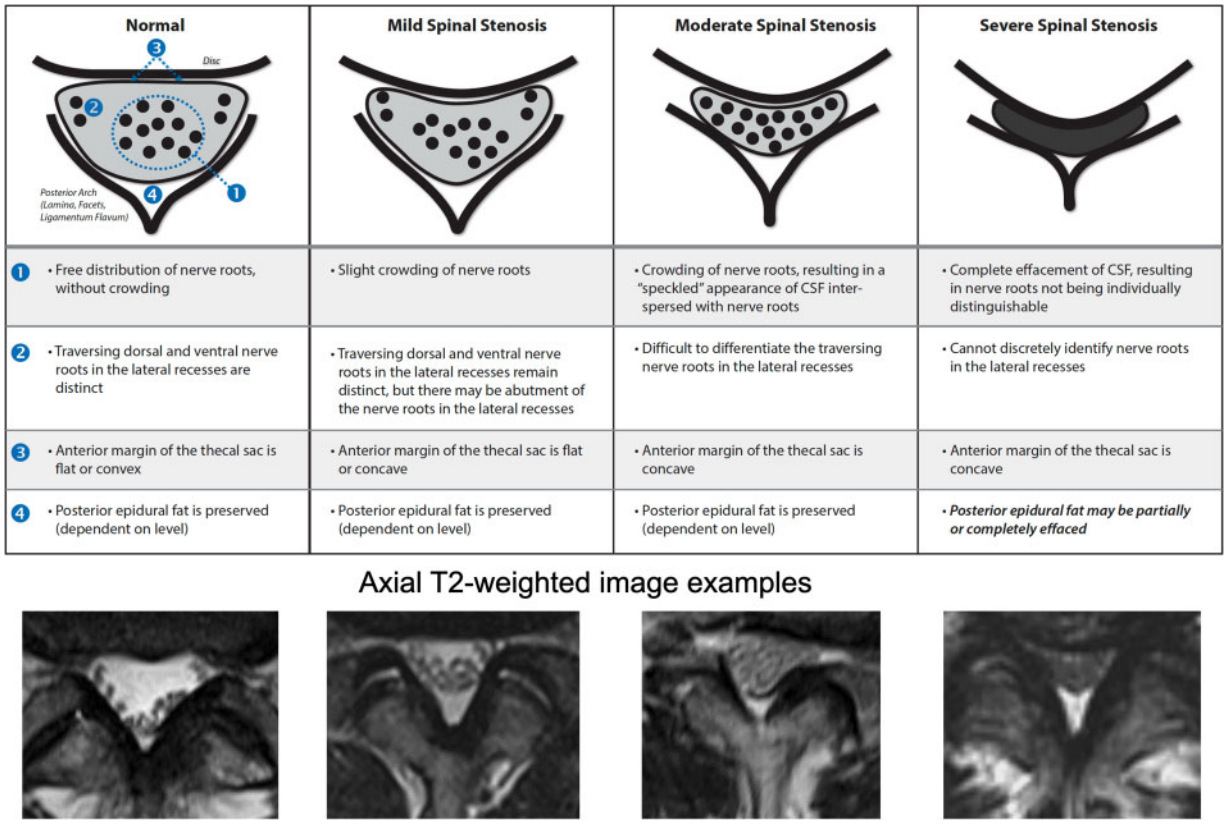


Figure 1. Spinal canal stenosis classification.

foraminal stenosis with perineural fat effacement in four directions but without nerve compressions, and grade 3 being severe stenosis with nerve root collapse or morphological change. In our clinical practice, we have sometimes found it challenging to differentiate between moderate and severe stenosis as originally described. Therefore, our grading of foraminal stenosis (Figure 2) is as follows, performed on sagittal T1-weighted images:

Normal (no foraminal stenosis): Nerve roots freely exit neural foramen, surrounded circumferentially by epidural fat. Mild narrowing of the inferior of the foraminal "keyhole" is still considered normal, provided there is preserved epidural fat surrounding the nerve.

Mild foraminal stenosis: Nerve root is in contact with the walls of the neural foramen in two planes (either anterior–posterior or cranial–caudal) or in three planes, but the remainder of the epidural fat is preserved, and the nerve root is normal in morphology.

Moderate foraminal stenosis: There is contact of the nerve root on at least two sides *and* substantial or near-complete effacement of epidural fat throughout the foramen. The nerve root remains normal in morphology.

Severe foraminal stenosis: There is compression of the exiting nerve root or complete effacement of epidural fat.

Lateral Recess Stenosis

Lateral recess stenosis was assessed with a modified grading system developed by Pfirrman et al. [20], who described a four-grade scale based on compromise of the traversing nerve root (no compromise, contact of nerve root, deviation of nerve root, and compression of nerve root). We have simplified this classification into a three-grade scale performed on axial T2-weighted images (Figure 3), including normal, contact of the nerve root without compression, and compression of the nerve root.

Facet Arthropathy

Facet arthropathy was assessed with a combination of the MRI-based system described by Fujiwara et al. [34] and the combined MRI- and computed tomography–based system of Weishaupt et al. [35], with normal, mild, moderate, and severe, as described below and as shown in Figure 4. Intermediate grades of mild–moderate and moderate–severe are allowed but not explicitly described. Assessment was performed by using axial T2-weighted images.

Mild facet arthropathy: There are minimal to small osteophytes, without mass effect on the nerve roots or spinal canal. There can be mild subchondral irregularity or small subchondral cysts.

Moderate facet arthropathy: Any combination of moderate-sized osteophytes, moderate subchondral irregularly, or moderate-sized subchondral cysts or synovial cysts. There can be mild or moderate mass effect from the hypertrophic change. Effusion, periarticular edema, or osseous edema may be present.

Downloaded from https://academic.oup.com/painmedicine/article/22/7/1485/6169540 by Harvard Law School Library user on 11 November 2021

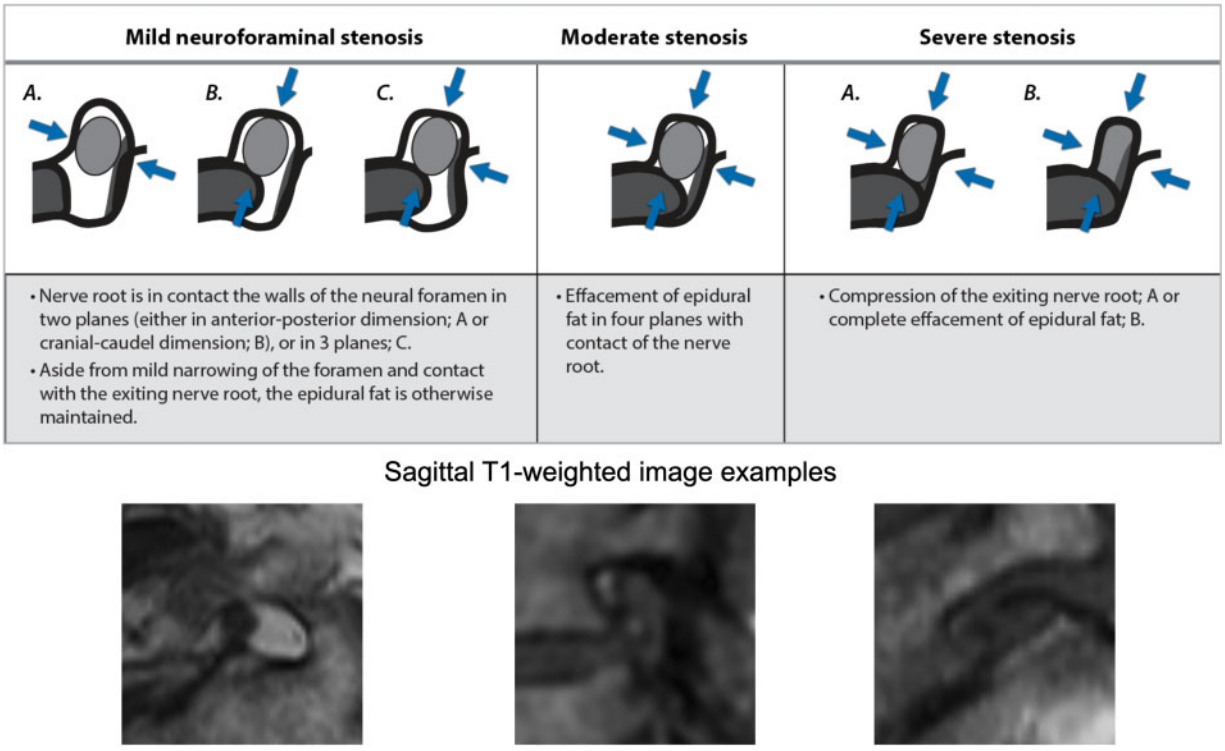


Figure 2. Foraminal stenosis classification.

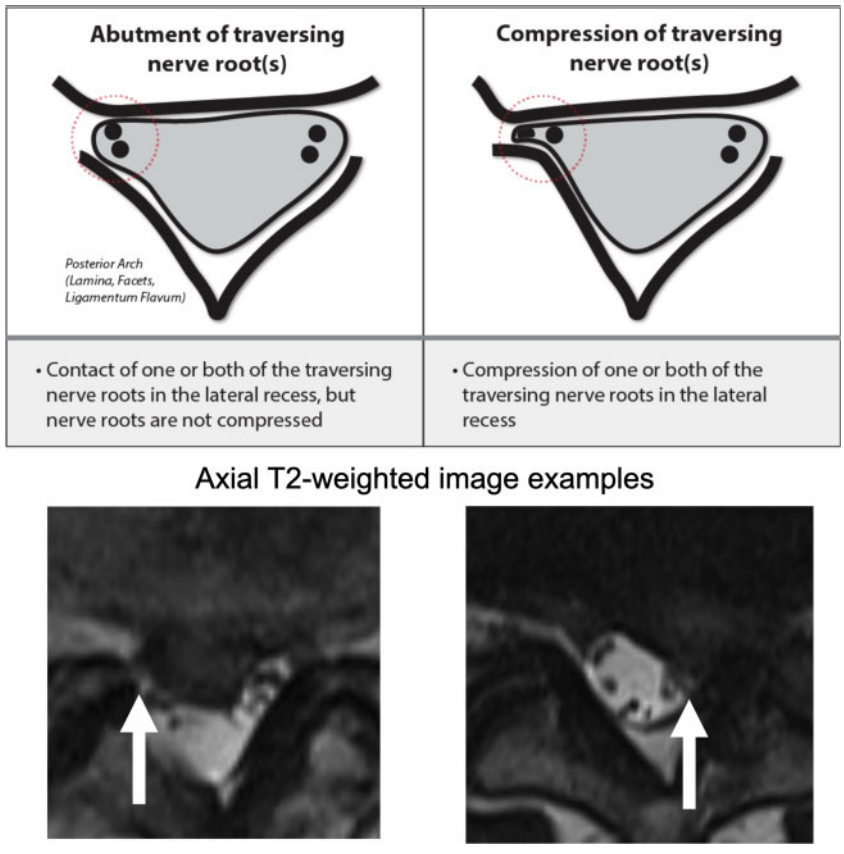


Figure 3. Lateral recess stenosis classification.

Downloaded from https://academic.oup.com/painmedicine/article/22/7/1485/6169540 by Harvard Law School Library user on 11 November 2021

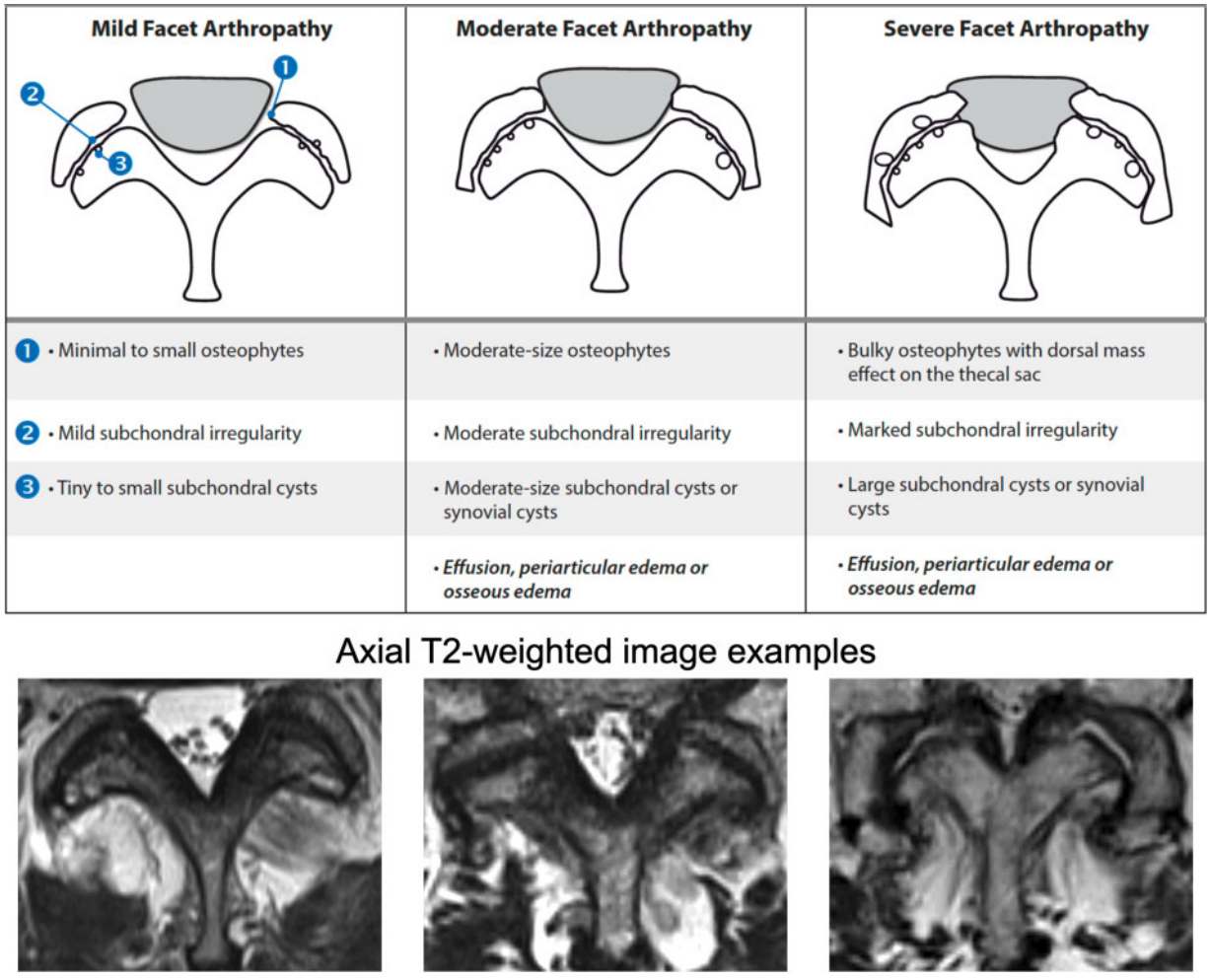


Figure 4. Facet arthropathy classification.

Severe facet arthropathy: There are bulky osteophytes with mass effect on the dorsal aspect of the thecal sac. Marked subchondral irregularity may be present. There can be large subchondral cysts or synovial cysts. Effusion, periarticular edema, or osseous edema may be present.

Patient Selection and Magnetic Resonance Scanning

This Health Insurance Portability and Accountability Act (HIPAA)–compliant study utilizing retrospective data received Institutional Review Board approval, with waiver of informed consent. The picture archiving and communication system was searched for 50 consecutive patients who underwent non-contrast MRI of the lumbar spine for chronic low back pain, radiculopathy, or symptoms of spinal stenosis. Examinations performed for concern for tumor, infection, or inflammatory conditions were excluded, as were patients with prior lumbar surgery.

All examinations were performed on one of 10 MRI scanners (five at 1.5 T and five at 3.0 T), manufactured by GE (Milwaukee, WI, USA) or Siemens (Erlangen, Germany). Each MRI included localizer, axial T1 and T2, and sagittal T1, T2, and short tau inversion recovery (STIR). Axial images were performed with a slice thickness of 4 mm, with a slice spacing of 5 mm. Sagittal images were performed with a slice thickness of 4 mm, with a slice spacing of 4.52 mm.

Reading Sessions and Classification

The three readers included the neurosurgeon with 5 years of experience, the orthopedic surgeon with 2 years of experience, and the physiatrist with 18 years of experience. All three readers helped create the classification and were also provided with the PowerPoint presentation, video, and single-page summary. All readers re-reviewed this material before the reading sessions, despite familiarity with the grading scheme. All reading sessions were independently performed on de-identified Digital Imaging and Communications in Medicine (DICOM)–formatted images on a secure Web-based picture archiving and communication system that integrates data collection with data display [36]. The readers were instructed on how to use the Web-based system before interpreting the studies. All readers were blinded to all identifying patient health data.

Downloaded from https://academic.oup.com/painmedicine/article/22/7/1485/6169540 by Harvard Law School Library user on 11 November 2021

For each case, the readers assessed the four parameters in the classification at the L4–L5 and L5–S1 levels and provided separate classifications for right and left foramina and lateral recesses at each level. Spinal stenosis, foraminal stenosis, and facet arthropathy were each assessed as one of six grades: normal, mild, mild–moderate, moderate, moderate–severe, and severe. Lateral recess stenosis was assessed as one of three grades: normal, abutment of nerve root, or compression of nerve root. Grading was performed with a multiple-choice radio button interface on the online platform, and results were transferred into Microsoft Excel to prepare for data analysis.

Data Analysis

Statistical analysis was performed in R 3.4.1 (R Foundation for Statistical Computing, Vienna, Austria). The prevalence of each severity rating for each of the four parameters was determined by summing each grade assignment divided by the total number of assignments for the given parameter, including all readers. The four parameters were assessed for inter-observer agreement among the three readers through the use of the weighted Cohen's kappa coefficient. This test compares agreement between two observers; assessment of reader agreement was performed by calculating a single kappa value with 95% confidence intervals for a matrix consisting of the three paired responses (A–B, A–C, and B–C pairs concatenated into two columns consisting of AAB and BCC). Cohen's kappa coefficient was interpreted as per the standard of Landis and Koch [37]: kappa <0 is poor agreement; 0.00–0.20, slight agreement; 0.21–0.40, fair agreement; 0.41–0.60, moderate agreement; 0.61–0.80, substantial agreement; and 0.81–1.00, almost perfect agreement.

Results

Each of the three readers evaluated 100 unique levels for spinal stenosis (L4–L5 and L5–S1 for each of the 50 patients, yielding a total of 300 levels evaluated among all readers). Of those, 194 were assigned no stenosis (65%), 58 were assigned mild (19%), 7 were assigned mild–moderate (2%), 14 were assigned moderate (5%), 15 were assigned moderate–severe (5%), and 12 were assigned severe (4%). Each of the three readers assigned 200 gradings of neural foraminal stenosis (for a total of 600 gradings). Of those, 360 were assigned no stenosis (60%), 132 were assigned mild (22%), 22 were assigned mild–moderate (4%), 39 were assigned moderate (6%), 15 were assigned moderate–severe (3%), and 32 were assigned severe (5%). Each of the three readers assigned 200 gradings of lateral recess stenosis (for a total of 600 gradings). Of those, 323 were assigned no stenosis (48%), 198 were assigned nerve root abutment (29%), and 158 were assigned nerve root compression (23%). Each of the three readers assigned 100 gradings of facet arthropathy (for a total of 300 gradings). Of those, 61 were assigned no stenosis (20%), 135 were assigned mild (45%), 30 were assigned mild–moderate (10%), 39 were assigned moderate (13%), 14 were assigned moderate–severe (5%), and 21 were assigned severe (7%). The prevalence of each grading as determined by the readers is shown in Figure 5.

For spinal stenosis, the readers demonstrated substantial agreement ($\kappa = 0.702$), with a 95% confidence interval (CI) of 0.641–0.764. The percentage of cases with exact classification was 56%, which rose to 91% for assessments within one grade (within the six-point grading scale including intermediate grades). For foraminal stenosis, the readers demonstrated moderate agreement ($\kappa = 0.544$), with a 95% CI of 0.486–0.602. The percentage of cases with exact classification was 41.5%, which rose to 78.5% for assessments within one grade (within the six-point scale). For lateral recess stenosis, the readers demonstrated fair agreement ($\kappa = 0.323$), with a 95% CI of 0.255–0.392. The percentage of cases with exact classification was 34%, which rose to 90.5% for assessments within one grade (within the three-point scale). For facet arthropathy, the readers demonstrated moderate agreement ($\kappa = 0.557$), with a 95% CI of 0.495–0.620. The percentage of cases with exact classification was 33%, which rose to 76% for assessments within one grade (within the six-point scale).

The summary of reader agreement across the four parameters is shown in Figure 6. Representative images are shown in Figures 7–9.

Discussion

Through the modification and simplification of previously reported systems, we have described a collaborative multidisciplinary classification for comprehensive lumbar spine MRI reporting. The goals of this classification scheme are to describe pertinent findings to guide potential interventional management and to aid cross-specialty communication and reporting of degenerative spine MRI findings. Using the classification, three non-radiologist spine experts from different specialties reported substantial agreement for spinal stenosis, moderate agreement for foraminal stenosis and facet arthropathy, and fair agreement for lateral recess stenosis. There have been multiple studies reporting various classification schemes and inter-reader agreement on individual or several parameters, as summarized in Table 1. However, to our knowledge, ours is the first study to propose and evaluate a *comprehensive* assessment and the first to include non-radiologists from three different specialties as the readers.

Each parameter evaluated was chosen because specific surgical or interventional pain procedures could be targeted to each site. For instance, laminotomy/laminectomy or facetectomy could surgically treat spinal

Downloaded from https://academic.oup.com/painmedicine/article/22/7/1485/6169540 by Harvard Law School Library user on 11 November 2021

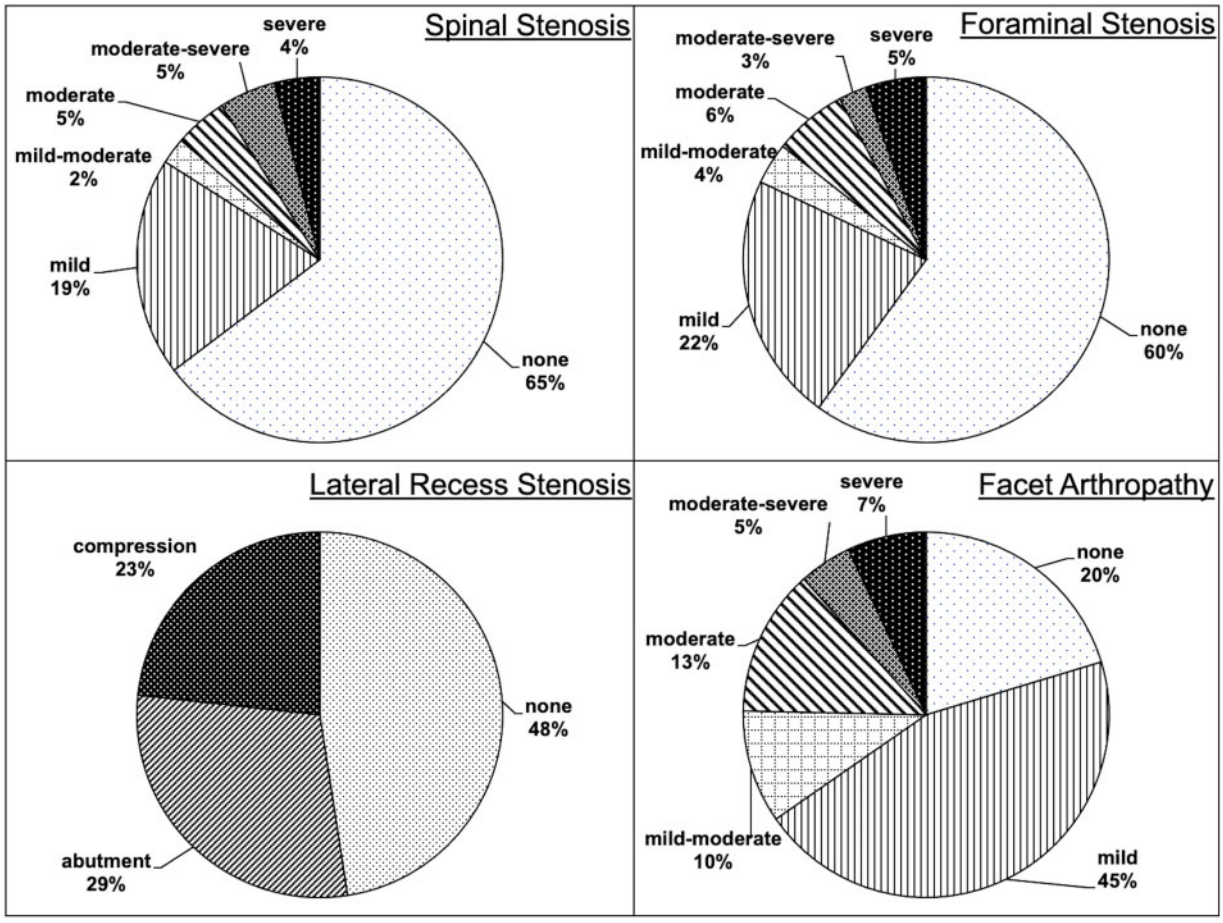


Figure 5. Prevalence of severity rating across all four parameters based off the simplified classification system of lumbar spine degeneration.

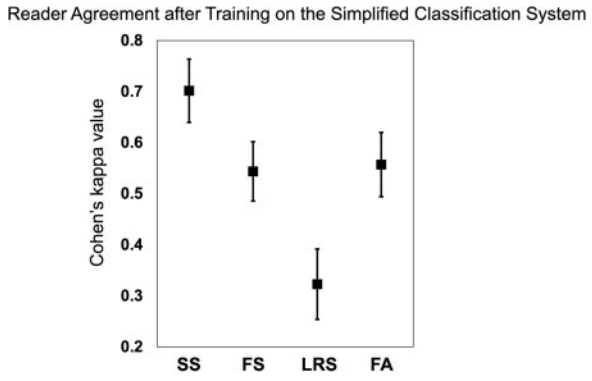


Figure 6. Plot demonstrating inter-reader agreement across the four parameters with 95% confidence intervals. SS=spinal stenosis; FS=foraminal stenosis; LRS=lateral recess stenosis; FA=facet arthropathy.

stenosis, lateral recess stenosis, or foraminal stenosis. Additionally, delineation of the specific site of nerve root impingement can have important implications for injection procedures. For instance, a nerve root impinged within the foramen could be targeted by a transforaminal epidural steroid injection at the site of stenosis, whereas that same level nerve root impinged in the lateral recess may be best targeted with a preganglionic approach at the level above [38]. Additionally, the presence of facet arthropathy may alter the surgical approach or can be individually targeted with an intra-articular facet injection or medial branch block.

For spinal stenosis, Schizas [33] described an inter-reader agreement of 0.44 for their eight-grade (including normal) classification among four readers (including one radiologist, one spine surgeon, and two junior orthopedic surgeons). Using a modification of Schizas' system, Weber et al. [39] described a range of paired inter-reader agreement for assessment of spinal stenosis of 0.58–0.76 among two neurosurgeons and two radiologists. Other reports have described inter-reader agreement between radiologists of 0.73–0.953 [40], between orthopedic surgeons of 0.681 [41], and between radiologists and neurosurgeons of 0.33–0.4 [42]. Our results of an inter-reader agreement of $\kappa = 0.702$ for spinal stenosis fall within the reported range, higher than the reported results from Schizas, Al-Tameemi, and Yuan and slightly below the reported results of Guen.

For evaluation of foraminal stenosis, Lee et al. [24] reported an inter-reader agreement between two radiologist readers of 0.800–0.957, varying slightly depending

Downloaded from https://academic.oup.com/painmedicine/article/22/7/1485/6169540 by Harvard Law School Library user on 11 November 2021

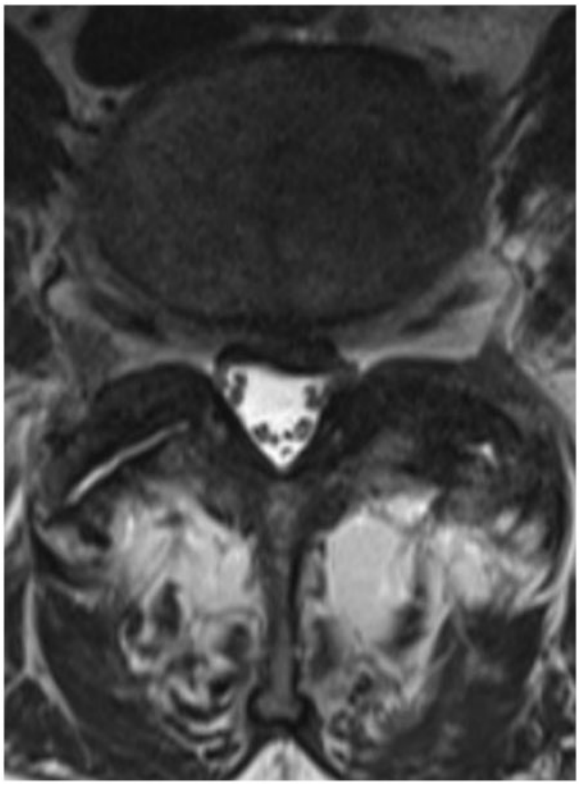


Figure 7. Axial T2-weighted image at the L4–L5 level. All three readers described the spinal canal as mildly stenosed and the left lateral recess as demonstrating nerve root compression. Two readers describe degree of facet arthropathy as moderate-to-severe, and one described it as severe.

on the level and side. Park et al. [22] reported an agreement of 0.734 with the Wildermuth system [43] and 0.767 with the aforementioned classification scheme proposed by Lee et al. Our results of moderate agreement ($\kappa = 0.544$) were lower than the results of Lee and Park.

For lateral recess stenosis, Pfirmann et al. [20] reported an inter-reader agreement of 0.62–0.67 among three readers including one spine surgeon and two radiologists. Kaliya-Perumal et al. [29] more recently reported an inter-reader agreement of 0.521 among three orthopedic surgery residents. Our results of fair agreement ($\kappa = 0.323$) for evaluation of lateral recess stenosis were lower than these previously reported results.

For facet arthropathy, Weishaupt et al. [35] reported inter-reader agreement of 0.41 for their magnetic resonance–based classification system between two radiologists. Fujiwara et al. [34] reported inter-reader agreement of 0.636 between two orthopedists. Berg et al. [28] reported inter-reader agreement of 0.20–0.32 among three radiologists using the Weishaupt classification system. Stieber et al. [27] reported an agreement of 0.21 among 10 spine surgery attendings and an agreement of 0.07 among three spine surgery fellows using the Fujiwara classification system. Our inter-reader agreement of $\kappa = 0.557$ was overall higher than most of these previously reported studies.



Figure 8. Axial T2-weighted image at the L4–L5 level in a patient with grade 1 anterolisthesis of L4 on L5, with no spondylolysis. Two readers described spinal canal stenosis as moderate, and one described it as moderate-to-severe. All three readers described the facet arthropathy as severe.

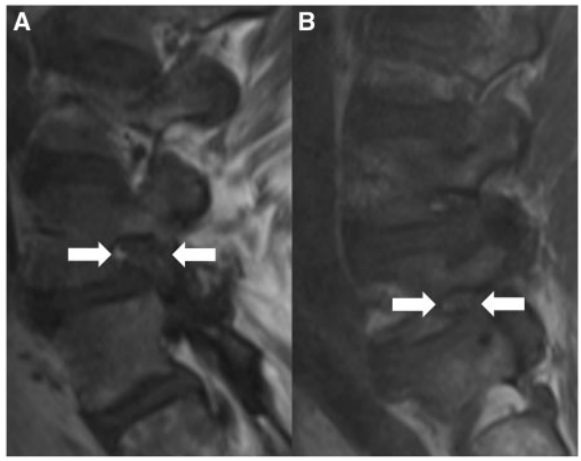


Figure 9. Sagittal T1-weighted images from two different patients. **(A)** All three readers described the right L4–5 foraminal stenosis as moderate. **(B)** All three readers described the right L5–S1 foraminal stenosis as severe.

Overall, our multi-reader agreement was within the reported range for spinal stenosis, lower than previously reported studies for foraminal and lateral recess stenosis, and higher than reports for facet arthropathy.

Downloaded from https://academic.oup.com/painmedicine/article/22/7/1485/6169540 by Harvard Law School Library user on 11 November 2021

Table 1. Comparison of previously published classification systems of lumbar degeneration on MRI

Parameter/Study	Classification	Lumbar Level	Inter-Reader Agreement (Kappa Value)	Intra-Reader Agreement (Kappa Value)	Readers
Spinal stenosis					
Schizas et al. (2010)	Original	Every	0.44	0.65	1 radiologist 1 spine surgeon 2 junior orthopedists
Weber et al. (2015)	Modified from Schizas et al. (2010)	L2–L3 L3–L4 L4–L5	Neurosurgeon 1 and 2: 0.76 Radiologist 1 and 2: 0.65 Neurosurgeon 1 and Radiologist 1: 0.63 Neurosurgeon 1 and Radiologist 2: 0.67 Neurosurgeon 2 and Radiologist 1: 0.58 Neurosurgeon 2 and Radiologist 2: 0.59	Neurosurgeon 1: 0.96 Neurosurgeon 2: 0.95 Radiologist 1: 0.78 Radiologist 2: 0.81	2 neurosurgeons 2 radiologists
Guen et al. (2011)	Original	Every	0.730–0.953	0.863–0.900	4 radiologists
Yuan et al. (2016)	Original	Every	0.681	0.818	3 junior orthopedists
Al-Tameemi et al. (2017)	Guen et al. (2011)	Every	Radiologists: 0.4 Radiologist-Neurosurgeon: 0.33	N/A	2 radiologists 1 neurosurgeon
Foraminal stenosis					
Lee et al. (2010)	Original	Every	L4–L5: right, 0.929; left, 0.942 L5–S1: right 0.919; left, 0.909	L4–L5: right, 0.957; left, 0.885 L5–S1: right, 0.800; left, 0.905	2 radiologists
Park et al. (2012)	Lee et al. (2010) Wildermuth et al. (1998)	L2–L3 L3–L4 L4–L5 L5–S1	Lee system, 0.767 Wildermuth system, 0.734	N/A	2 radiologists
Lateral recess stenosis					
Pfirrmann et al. (2004)	Original	Every	0.62–0.67	0.72–0.77	1 spine surgeon 2 radiologists
Kaliya-Perumal et al. (2018)	Pfirrmann et al. (2009)	L2–L3 L3–L4 L4–L5 L5–S1	0.521	0.696	3 orthopedic residents
Facet arthropathy					
Weishaupt et al. (1999)	Original	Every	0.60 for CT 0.41 for MRI	Radiologist 1: 0.70 for CT; 0.70 for MR Radiologist 2: 0.77 for CT; 0.76 for MR	2 radiologists
Fujiwara et al. (1999)	Original	L3–L4 L4–L5 L5–S1	0.636	N/A	2 orthopedists
Stieber et al. (2009)	Fujiwara et al. (1999) Pathria et al. (1987)	L3–L4 L4–L5 L5–S1	Attendings: 0.21 Fellows: 0.07	Attendings: 0.36 Fellows: 0.26	10 spine surgeons 3 spine surgery fellows
Berg et al. (2019)	Weishaupt et al. (1999)	L3–L4 L4–L5 L5–S1	0.20–0.32	N/A	3 radiologists

CT=computed tomography.

Somewhat unintuitively, out of the four parameters, we found the lowest agreement in the assessment of lateral recess stenosis, despite the fact that the ordinal scale was actually the smallest of the four parameters, which one would suspect should promote the most concordance. We suspect that this may be related to disagreement among readers reporting a normal lateral recess vs nerve root abutment; at the lower lumbar spine levels, even a small bulge may appear to abut the traversing nerve roots. This speculation is further substantiated by the dramatic increase in percentage agreement when the percentage with exact-grade agreement (34%) is compared with the percentage with within-one-grade agreement (90.5%), which is the largest such increase across the four parameters.

Downloaded from https://academic.oup.com/painmedicine/article/22/7/1485/6169540 by Harvard Law School Library user on 11 November 2021

Limitations

This study has several important limitations. It is a single-institution study, with the classification and reading performed by a group of spine physicians who have a collaborative working relationship. As a result, it is possible that reader variability is underestimated, although the actual impact of this is likely very low or even counteracted by the fact that all readers have disparate training backgrounds at different institutions in different specialties. In fact, the different specialties of the readers constitute perhaps one of the unique aspects of this study, as they demonstrate the cross-disciplinary potential of the classification system. However, future studies can seek to validate these findings in a multi-institutional study to eliminate this potential bias. There are also relatively few readers, with three independent readers. However, as the brief literature review demonstrated, the majority of other studies have used either two or three readers, and relatively few have used a larger number of readers.

Another important limitation of the study is that a comparison was not performed, i.e., with and without the proposed classification. Such a study design would not have been possible with this group of readers, as the readers were intimately familiar with and instrumental in creating the classification system. Additionally, though aiming to be comprehensive, the classification does not include many common findings of spine degeneration because many of them do not often change clinical or interventional management. For example, ligamentum flavum thickening may contribute to spinal stenosis, but it is not an entity that is treated in isolation. Of course, the presence and morphology of disc abnormalities can be of critical importance, but the terminology and description of disc pathology is a complex topic that has been thoroughly described previously.

Future work will assess variability within and between radiology subspecialties, as well as between radiology and non-radiology spine experts, as radiologists play an important role in the interpretation and communication of these findings. Variability in the interpretation of spine MRI is well-known issue [13–15], and if validated, this comprehensive simplified classification system may assist in improving concordance in lumbar spine MRI interpretation to improve care of this large patient population.

Conclusion

A simplified universal grading system for four key degenerative lumbar spine MRI findings shows substantial agreement among non-radiologist expert spine physicians in the assessment of spinal stenosis, moderate agreement for foraminal stenosis and facet arthropathy, and fair agreement for lateral recess stenosis. These definitions may be useful in everyday clinical practice and aid cross-specialty communication and reporting consistency.

References

1. Hart LG, Deyo RA, Cherkin DC. Physician office visits for low back pain. Frequency, clinical evaluation, and treatment patterns from a U.S. national survey. Spine 1995;20(1):11–9.
2. Licciardone JC. The epidemiology and medical management of low back pain during ambulatory medical care visits in the United States. Osteopath Med Prim Care 2008;2:11.
3. St Sauver JL, Warner DO, Yawn BP, et al. Why patients visit their doctors: Assessing the most prevalent conditions in a defined American population. Mayo Clin Proc 2013;88(1):56–67.
4. Martin BI, Deyo RA, Mirza SK, et al. Expenditures and health status among adults with back and neck problems. JAMA 2008; 299(6):656–64.
5. Dagenais S, Caro J, Haldeman S. A systematic review of low back pain cost of illness studies in the United States and internationally. Spine J 2008;8(1):8–20.
6. Boden SD, Davis DO, Dina TS, Patronas NJ, Wiesel SW. Abnormal magnetic-resonance scans of the lumbar spine in asymptomatic subjects. A prospective investigation. J Bone Joint Surg Am 1990;72(3):403–8.
7. Jensen MC, Brant-Zawadzki MN, Obuchowski N, et al. Magnetic resonance imaging of the lumbar spine in people without back pain. N Engl J Med 1994;331(2):69–73.
8. Weishaupt D, Zanetti M, Hodler J, Boos N. MR imaging of the lumbar spine: Prevalence of intervertebral disk extrusion and sequestration, nerve root compression, end plate abnormalities, and osteoarthritis of the facet joints in asymptomatic volunteers. Radiology 1998;209(3):661–6.
9. Kalichman L, Li L, Kim DH, et al. Facet joint osteoarthritis and low back pain in the community-based population. Spine 2008; 33(23):2560–5.
10. Brinjikji W, Luetmer PH, Comstock B, et al. Systematic literature review of imaging features of spinal degeneration in asymptomatic populations. AJNR Am J Neuroradiol 2015;36 (4):811–6.
11. Steffens D, Hancock MJ, Maher CG, et al. Does magnetic resonance imaging predict future low back pain? A systematic review. Eur J Pain Lond Engl 2014;18(6):755–65.
12. Chou R, Qaseem A, Snow V, et al.; American Pain Society Low Back Pain Guidelines Panel. Diagnosis and treatment of low back pain: A joint clinical practice guideline from the American College of Physicians and the American Pain Society. Ann Intern Med 2007;147(7):478–91.
13. Fu MC, Buerba RA, Long WD, et al. Interrater and intrarater agreements of magnetic resonance imaging findings in the lumbar spine: Significant variability across degenerative conditions. Spine J 2014;14(10):2442–8.
14. Winklhofer S, Held U, Burgstaller JM, et al. Degenerative lumbar spinal canal stenosis: Intra- and inter-reader agreement for magnetic resonance imaging parameters. Eur Spine J 2017;26 (2):353–61.
15. Carrino JA, Lurie JD, Tosteson ANA, et al. Lumbar spine: Reliability of MR imaging findings. Radiology 2009;250 (1):161–70.
16. Speciale AC, Pietrobon R, Urban CW, et al. Observer variability in assessing lumbar spinal stenosis severity on magnetic resonance imaging and its relation to cross-sectional spinal canal area. Spine 2002;27(10):1082–6.
17. van Rijn JC, Klemetsö N, Reitsma JB, et al. Observer variation in MRI evaluation of patients suspected of lumbar disk herniation. AJR Am J Roentgenol 2005;184(1):299–303.
18. Miskin N, Gaviola GC, Huang RY, et al. Intra- and intersubspecialty variability in lumbar spine MRI interpretation: A

Downloaded from https://academic.oup.com/painmedicine/article/22/7/1485/6169540 by Harvard Law School Library user on 11 November 2021

multireader study comparing musculoskeletal radiologists and neuroradiologists. Curr Probl Diagn Radiol 2020;49(3):182–7.
19. Mulconrey DS, Knight RQ, Bramble JD, Paknikar S, Harty PA. Interobserver reliability in the interpretation of diagnostic lumbar MRI and nuclear imaging. Spine J 2006;6(2):177–84.
20. Pfirrmann CWA, Dora C, Schmid MR, et al. MR image-based grading of lumbar nerve root compromise due to disk herniation: Reliability study with surgical correlation. Radiology 2004;230 (2):583–8.
21. Steurer J, Roner S, Gnannt R, Hodler J, LumbSten Research Collaboration. Quantitative radiologic criteria for the diagnosis of lumbar spinal stenosis: A systematic literature review. BMC Musculoskelet Disord 2011;12(1):175.
22. Park H-J, Kim SS, Lee S-Y, et al. Clinical correlation of a new MR imaging method for assessing lumbar foraminal stenosis. AJNR Am J Neuroradiol 2012;33(5):818–22.
23. Park H-J, Kim SS, Lee Y-J, et al. Clinical correlation of a new practical MRI method for assessing central lumbar spinal stenosis. Br J Radiol 2013;86(1025):20120180.
24. Lee S, Lee JW, Yeom JS, et al. A practical MRI grading system for lumbar foraminal stenosis. AJR Am J Roentgenol 2010;194 (4):1095–8.
25. Modic MT. Modic type 1 and type 2 changes. J Neurosurg Spine 2007;6(2):150–1; discussion 151.
26. Jensen TS, Sorensen JS, Kjaer P. Intra- and interobserver reproducibility of vertebral endplate signal (Modic) changes in the lumbar spine: The Nordic Modic Consensus Group classification. Acta Radiol Stockh Swed 2007;48(7):748–54.
27. Stieber J, Quirno M, Cunningham M, Errico TJ, Bendo JA. The reliability of computed tomography and magnetic resonance imaging grading of lumbar facet arthropathy in total disc replacement patients. Spine 2009;34(23):E833–840.
28. Berg L, Thoresen H, Neckelmann G, et al. Facet arthropathy evaluation: CT or MRI? Eur Radiol 2019;29(9):4990–8.
29. Kaliya-Perumal A-K, Ariputhiran-Tamilselvam S-K, Luo C-A, et al. Revalidating Pfirrmann's magnetic resonance image-based grading of lumbar nerve root compromise by calculating reliability among orthopaedic residents. Clin Orthop Surg 2018;10 (2):210–5.
30. Pfirrmann CW, Metzdorf A, Zanetti M, Hodler J, Boos N. Magnetic resonance classification of lumbar intervertebral disc degeneration. Spine 2001;26(17):1873–8.
31. Griffith JF, Wang Y-XJ, Antonio GE, et al. Modified Pfirrmann grading system for lumbar intervertebral disc degeneration. Spine 2007;32(24):E708–712.
32. Yu L-P, Qian W-W, Yin G-Y, Ren Y-X, Hu Z-Y. MRI assessment of lumbar intervertebral disc degeneration with lumbar degenerative disease using the Pfirrmann grading systems. PloS One 2012;7(12):e48074.
33. Schizas C, Theumann N, Burn A, et al. Qualitative grading of severity of lumbar spinal stenosis based on the morphology of the dural sac on magnetic resonance images. Spine 2010;35(21):1919–24.
34. Fujiwara A, Tamai K, Yamato M, et al. The relationship between facet joint osteoarthritis and disc degeneration of the lumbar spine: An MRI study. Eur Spine 1999;8(5):396–401.
35. Weishaupt D, Zanetti M, Boos N, Hodler J. MR imaging and CT in osteoarthritis of the lumbar facet joints. Skeletal Radiol 1999;28(4):215–9.
36. Hostetter J, Khanna N, Mandell JC. Integration of a zero-footprint cloud-based picture archiving and communication system with customizable forms for radiology research and education. Acad Radiol 2018;25(6):811–8.
37. Landis JR, Koch GG. The measurement of observer agreement for categorical data. Biometrics 1977;33(1):159–74.
38. Jeong HS, Lee JW, Kim SH, et al. Effectiveness of transforaminal epidural steroid injection by using a preganglionic approach: A prospective randomized controlled study. Radiology 2007;245(2):584–90.
39. Weber C, Rao V, Gulati S, et al. Inter- and intraobserver agreement of morphological grading for central lumbar spinal stenosis on magnetic resonance imaging. Glob Spine J 2015;5(5):406–10.
40. Lee GY, Guen YL, Lee JW, et al. A new grading system of lumbar central canal stenosis on MRI: An easy and reliable method. Skeletal Radiol 2011;40(8):1033–9.
41. Yuan S, Zou Y, Li Y, Chen M, Yue Y. A clinically relevant MRI grading system for lumbar central canal stenosis. Clin Imaging 2016;40(6):1140–5.
42. Al-Tameemi HN, Al-Essawi S, Shukri M, Naji FK. Using magnetic resonance myelography to improve interobserver agreement in the evaluation of lumbar spinal canal stenosis and root compression. Asian Spine J 2017;11(2):198–203.
43. Wildermuth S, Zanetti M, Duewell S, et al. Lumbar spine: Quantitative and qualitative assessment of positional (upright flexion and extension) MR imaging and myelography. Radiology 1998;207(2):391–8.

Downloaded from https://academic.oup.com/painmedicine/article/22/7/1485/6169540 by Harvard Law School Library user on 11 November 2021